Global Renewable Energy in 2025: How Far Have We Come?

A data-driven overview of renewable energy's growing share in the global energy and electricity mix.

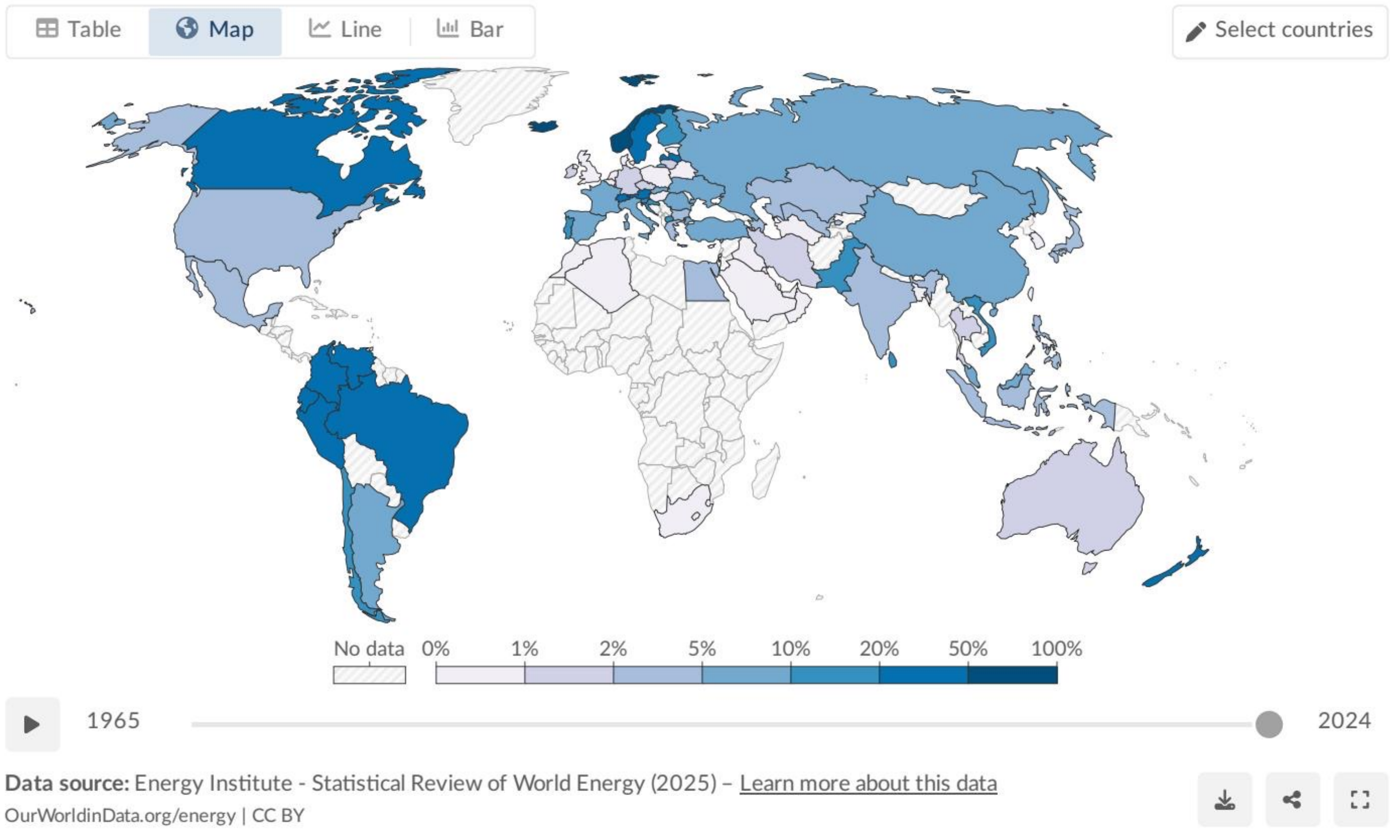
Source: Our World in Data

75% of Global GHG Emissions Still Come from Fossil Fuels

- Fossil fuels have dominated the global energy system since the Industrial Revolution.
- Three-quarters of global greenhouse gas emissions result from fossil fuel combustion for energy.
- Air pollution from fossil fuels causes at least 5 million premature deaths globally per year.
- Urgent action is required to rapidly shift toward low-carbon sources, including nuclear and renewable technologies.

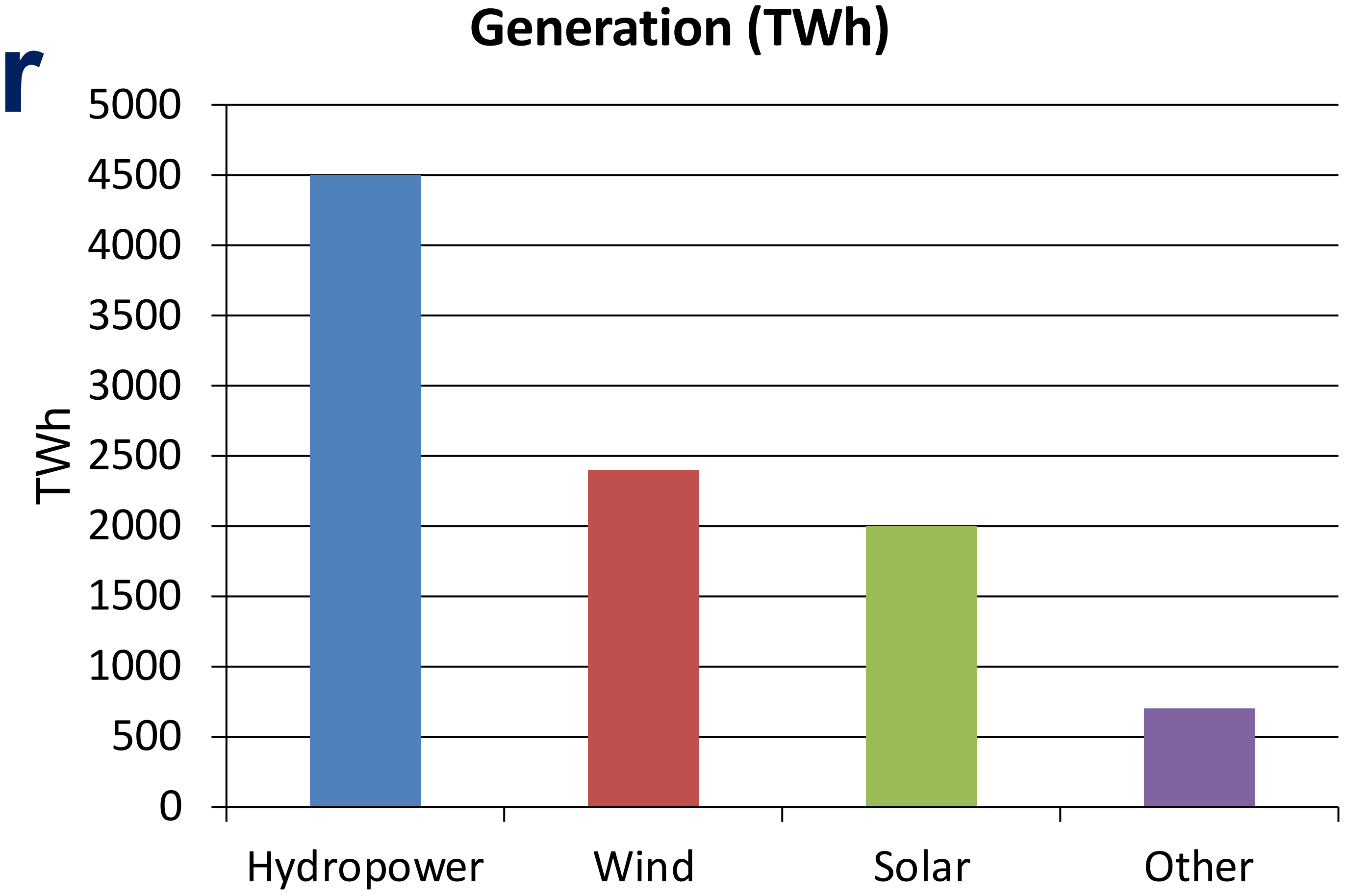
Renewables Now Supply ~14% of Global Primary Energ

Approximately one-seventh (~14%) of primary energy globally comes from renewables (2024, substitution method).
Primary energy includes electricity, transport, and heating.
Transport and heating are harder to decarbonize, leaving the primary energy share significantly lower than electricity alone.
Significant regional variation: Norway and Brazil lead, while Middle East and parts of Asia remain below 5%.



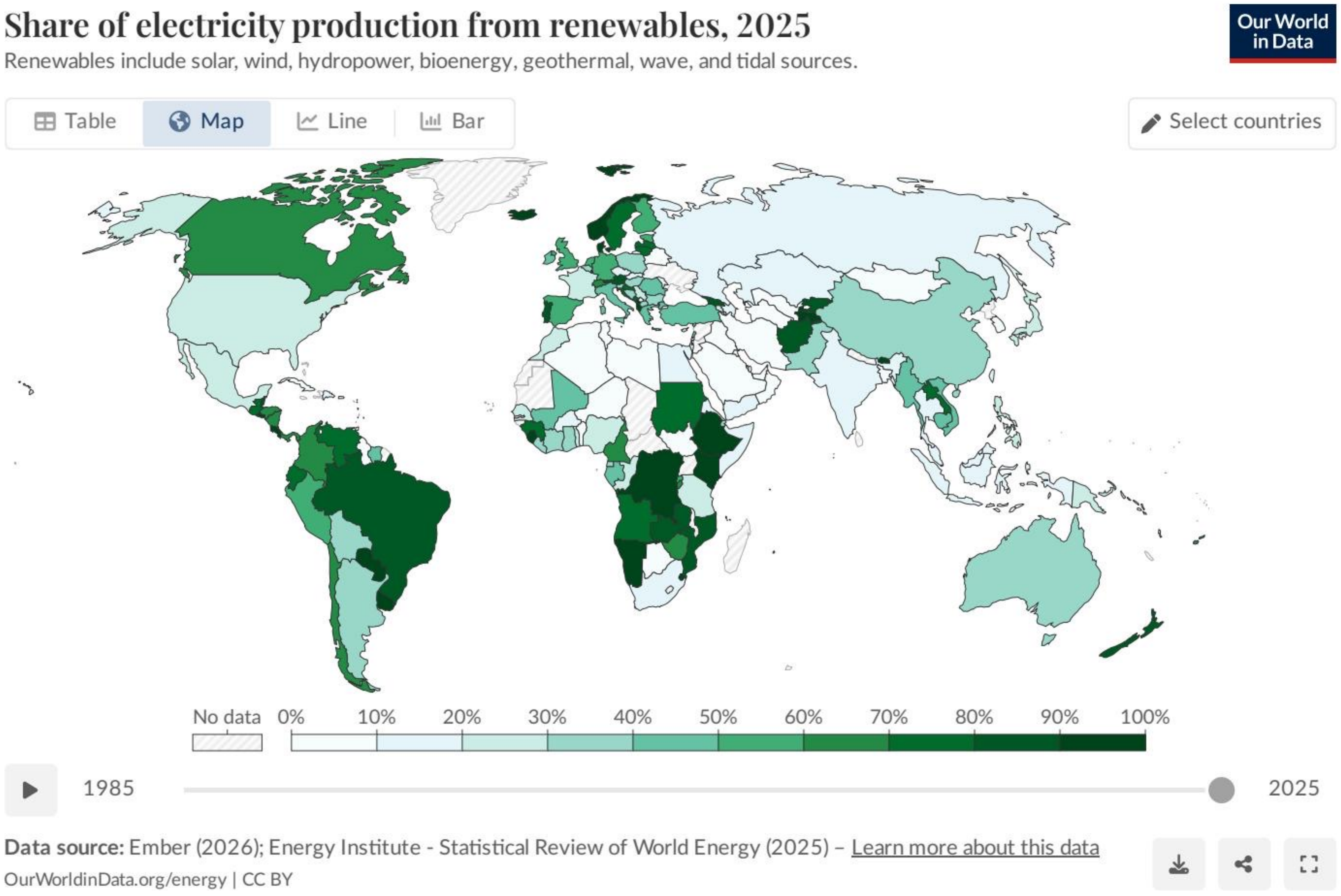
Renewable Electricity Generation Is Growing Rapidly, Led by Wind and Solar

Wind and solar are accelerating rapidly, driven by falling costs and policy support since ~2010.
Combined, wind and solar output now nearly matches total hydropower generation.
The total renewable generation mix is becoming increasingly diversified across technologies.



Almost One-Third of Global Electricity Now Comes from Rene

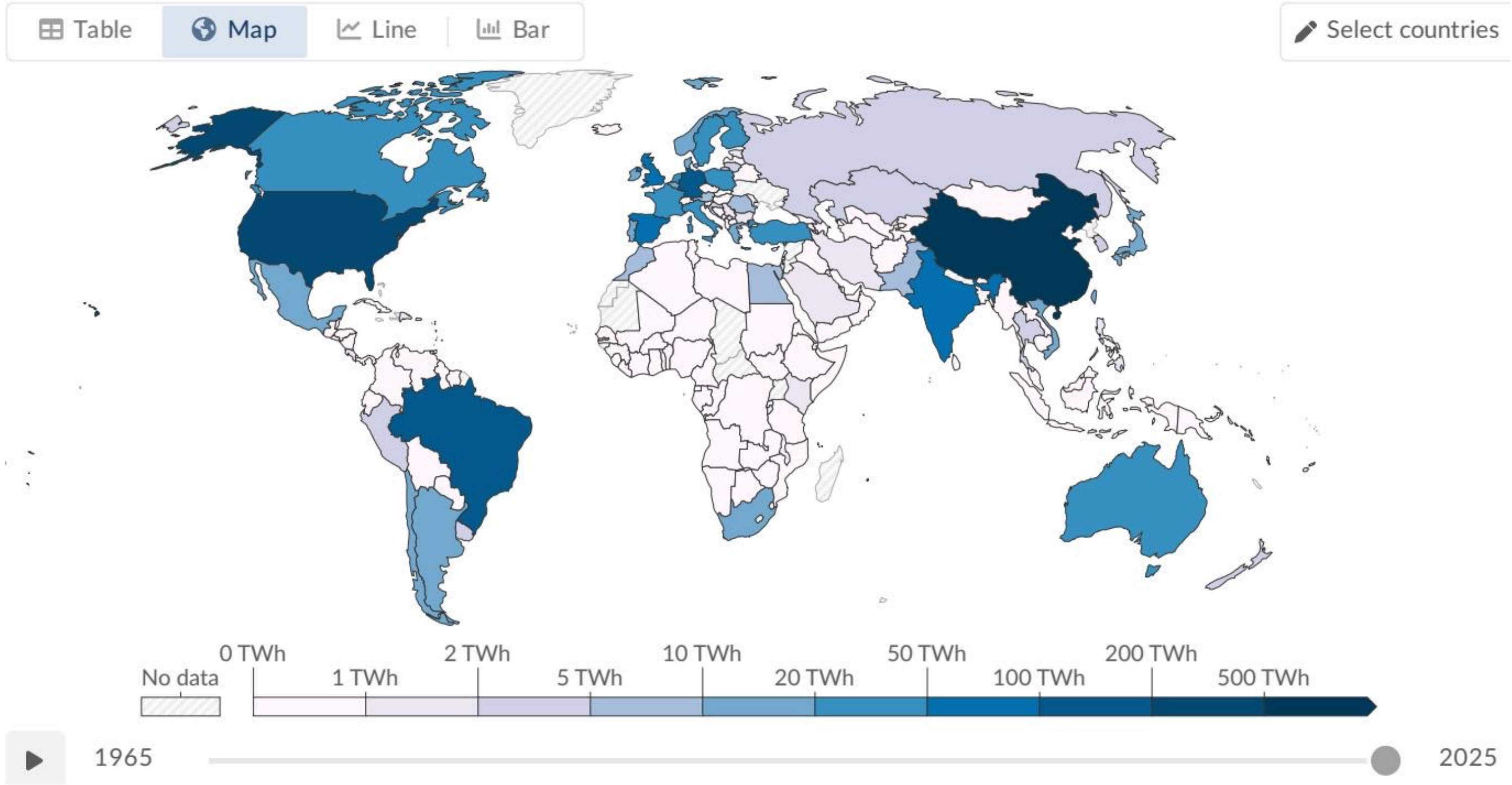
Country	Share
Norway	~98%
Brazil	~85%
Canada	~65%
China	~35%
India	~25%
Saudi Arabia	<2%



Renewable share in electricity (~30%) significantly outpaces primary energy.
Regional leaders benefit from large legacy hydro or aggressive wind/solar

Hydropower Remains the Largest Renewable Source at ~50% of

Hydropower generates ~4,500 TWh, approximately half of all renewable electricity.
It has provided scaled zero-carbon generation for over a century.
Top producers globally include China, Brazil, and Canada.
Growth potential is more constrained compared to wind and solar due to geographical limits.



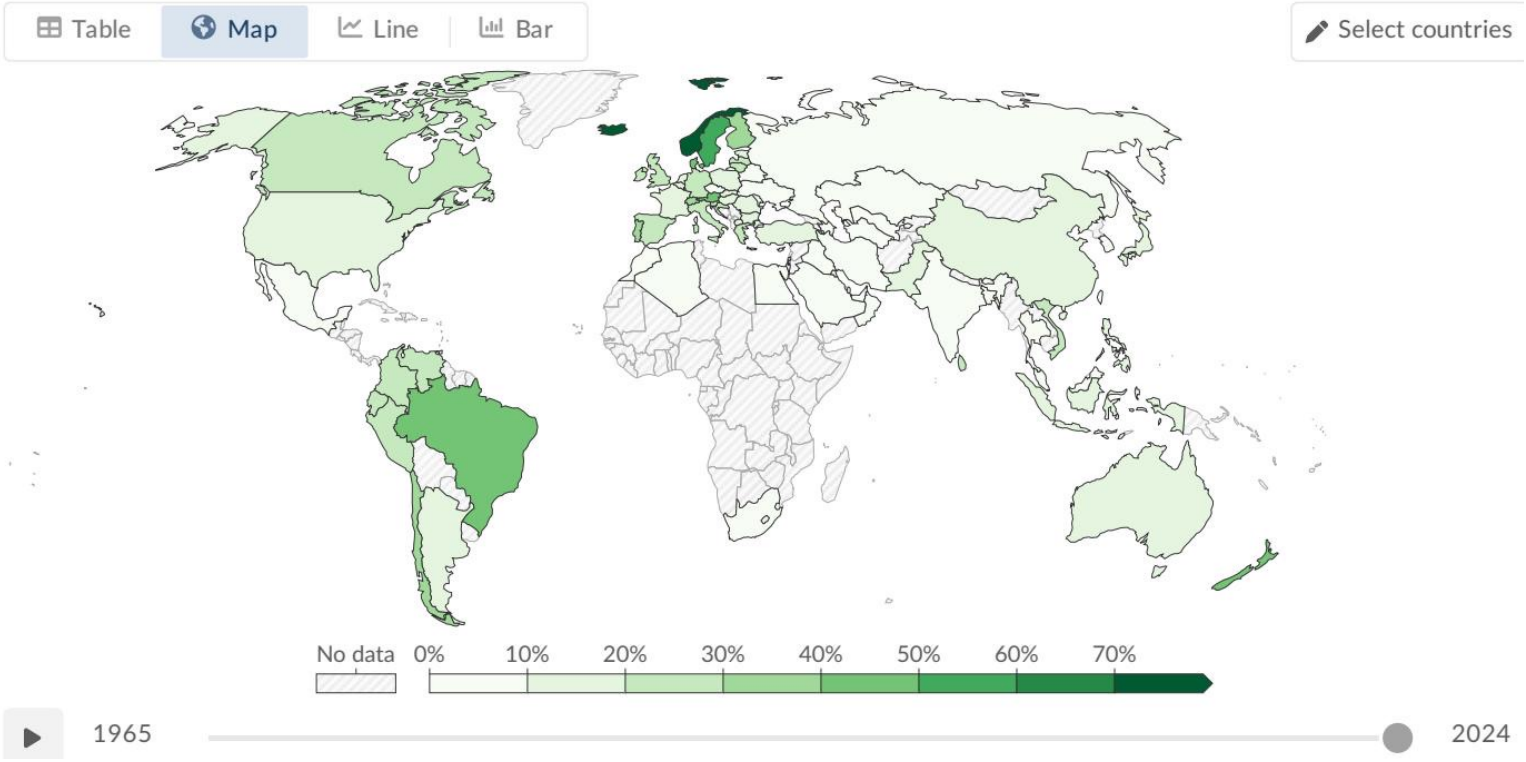
Wind and Solar Are the Fastest-Growing Energy Technologi

Wind Generation: ~2,400 TWh (~25% of renewable electricity).

Solar Generation: ~2,000 TWh (~21% of renewable electricity).

Solar is the single fastest-growing source globally, scaling dramatically since 2010.

Combined, wind and solar now produce nearly as much electricity as hydropower.



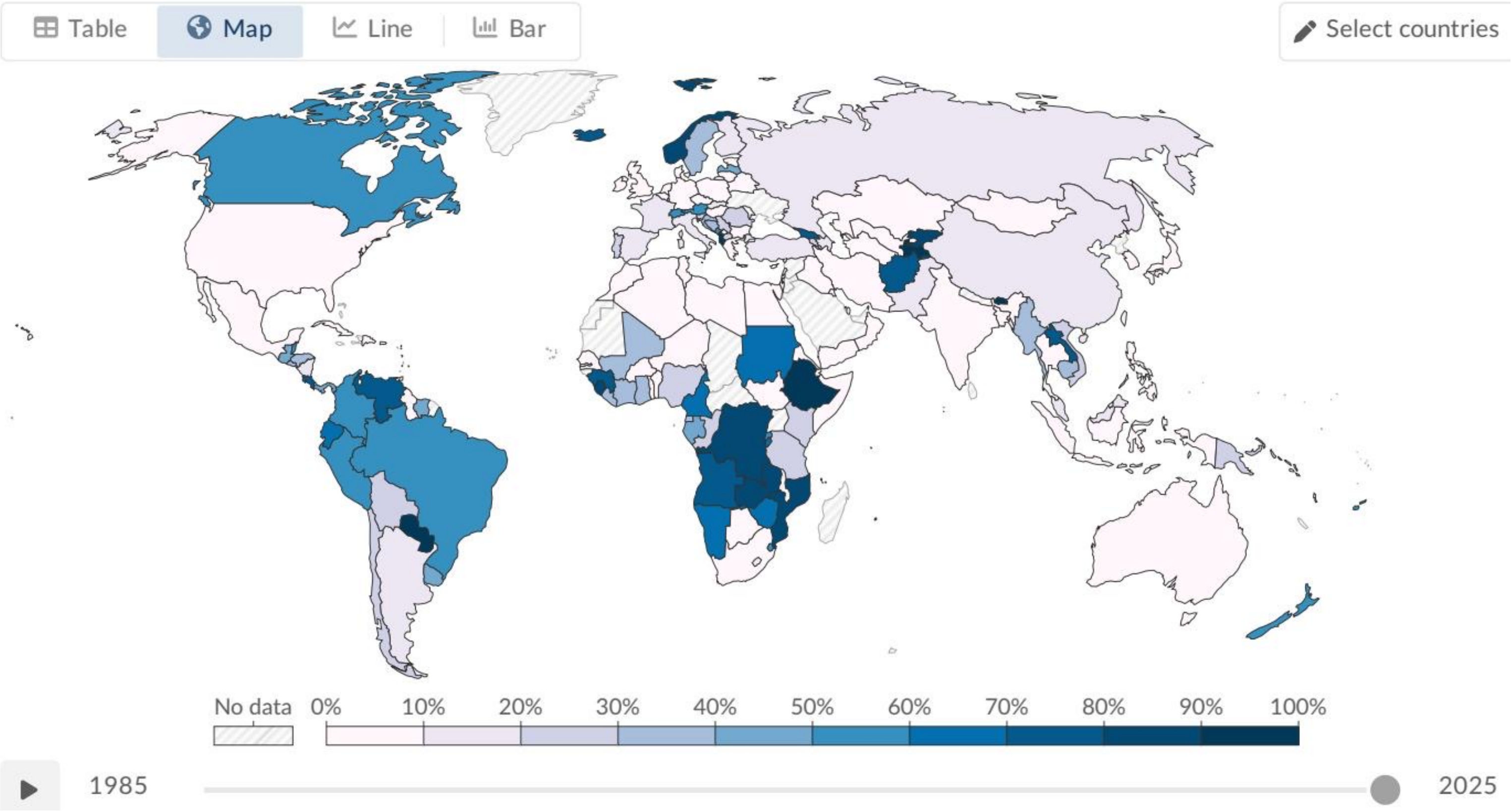
Stark Regional Gaps: Sub-Saharan Africa Lags While Scandinavia and Latin Ameri

Scandinavia, Brazil, and Canada lead globally with 65–98% renewables, largely from abundant hydropower.

European nations like Denmark and Germany lead in wind and solar deployment.

Much of Sub-Saharan Africa has minimal renewable electricity infrastructure, with shares below 10%.

Many Middle Eastern nations remain almost completely reliant on fossil fuels.



Key Takeaways: Renewables Are Growing Fast but the Transition Must Accelerate

- Progress Made: ~14% of global primary energy and ~30% of global electricity now come from renewables.
- Technology Shift: While hydropower dominates, wind and solar are the primary engines of rapid growth.
- Uneven Transition: Stark disparities persist. Leaders exceed 60-90% clean electricity; laggards remain below 10%.
- Sectoral Challenges: Transport and heating lag significantly behind electricity in decarbonizing.
- Future Outlook: To align with global climate targets, the deployment of wind and solar must dramatically accelerate while expanding infrastructure globally.